SidewalkPilot Steering Model Report

Combined current-label offline evaluation

Generated 2026-07-13 02:20:56. Every SidewalkPilot steering checkpoint (Series 1, 2, and 3) is evaluated on the SAME Series 3 dataset: 81,237 labeled field images. Series 1/2 (SteeringAutonomyV2, 200x66) and Series 3 (SidewalkPilotV3, 320x180) each run with their own preprocessing but are scored against the same steering labels, so the numbers are directly comparable. Checkpoint naming: SidewalkPilot-vX.Y.pth for final, SidewalkPilot-vX.Yb.pth for best.

Best class-balanced offline checkpoint: 3.2 (SidewalkPilot-v3.2.onnx) with balanced 9-bucket exact recall 34.0%, turn exact recall 31.1%, and turn within-one-bucket recall 57.7%. Second by the same ranking: 3.4 (SidewalkPilot-v3.4.onnx). Lowest MAE is 3.4b at 13.904 deg; MAE remains secondary because straight labels dominate this dataset.

Series 1 uses raw BGR and output scale 86. Series 2 uses output scale 85; v2.0/v2.0b use legacy HSV/CLAHE preprocessing, while v2.1 and newer use raw BGR. Series 3 is the heavy hybrid SidewalkPilotV3 at 320x180. All series are measured on the same Series 3 image set in servo degrees. Cross-series numbers are directly comparable for this image set, but Series 1/2 were trained under older data and calibration regimes. These are offline fit/transfer comparisons; field reliability still has to be proven on the car.

Series 3 Evaluation Sources

Source	Count	Purpose
D0702_16	50684	Series 3 July 2 base set: two manual field runs, 50,684 images
D0707_16	6524	Bright-sun HARD/diagonal shadows across the sidewalk (v3.2 shadow-robustness batch)
D0712_15	2102	Series 3 manual field collection, July 12 run 1
D0712_16	21927	Series 3 manual field collection, July 12 run 2 (103-frame bad segment removed)

Class-Balanced Model Ranking

Ranked by macro-average exact recall across the 9 steering buckets, then turn within-one-bucket recall, then MAE. This prevents the 66%-straight label majority from making a straight-collapsed model look best.

Rank	Model	Checkpoint filename	Prep	Bal9	Turn exact	Turn +/-1	ST exact	MAE	Med	Signed
1	3.2	SidewalkPilot-v3.2.onnx	raw BGR 320x180	34.0%	31.1%	57.7%	52.4%	16.78	9.65	-1.47
2	3.4	SidewalkPilot-v3.4.onnx	raw BGR 320x180	33.3%	32.6%	60.7%	65.5%	14.38	4.86	0.80
3	3.1	SidewalkPilot-v3.1.onnx	raw BGR 320x180	28.1%	26.8%	53.3%	55.2%	22.65	9.73	-2.35
4	3.1b	SidewalkPilot-v3.1b.onnx	raw BGR 320x180	27.4%	25.8%	52.3%	56.7%	20.96	9.59	-3.77
5	3.3	SidewalkPilot-v3.3.onnx	raw BGR 320x180	27.1%	21.2%	48.1%	68.5%	14.63	5.20	-5.35
6	3.4b	SidewalkPilot-v3.4b.onnx	raw BGR 320x180	25.6%	22.7%	52.2%	73.7%	13.90	2.68	-2.09
7	3.2b	SidewalkPilot-v3.2b.onnx	raw BGR 320x180	25.1%	19.8%	46.0%	67.2%	14.46	5.91	-4.69
8	3.3b	SidewalkPilot-v3.3b.onnx	raw BGR 320x180	19.2%	10.6%	36.5%	78.0%	16.46	2.18	-8.29
9	2.0	SidewalkPilot-v2.0.pth	HSV/CLAHE -> BGR	16.6%	11.8%	37.0%	48.1%	15.52	7.80	-6.04
10	2.0b	SidewalkPilot-v2.0b.pth	HSV/CLAHE -> BGR	16.6%	11.7%	37.0%	48.5%	15.48	7.74	-6.03
11	1.9	SidewalkPilot-v1.9.pth	raw BGR	16.4%	11.2%	36.2%	53.0%	15.07	7.33	-6.09

Rank	Model	Checkpoint filename	Prep	Bal9	Turn exact	Turn +/-1	ST exact	MAE	Med	Signed
12	1.9b	SidewalkPilot-v1.9b.pth	raw BGR	16.4%	11.2%	36.2%	53.3%	15.05	7.27	-6.18
13	2.1	SidewalkPilot-v2.1.pth	raw BGR	16.3%	12.8%	37.1%	36.7%	17.32	10.39	-6.55
14	2.1b	SidewalkPilot-v2.1b.pth	raw BGR	16.2%	12.7%	37.1%	36.9%	17.23	10.32	-6.44
15	3.0	SidewalkPilot-v3.0.onnx	raw BGR 320x180	16.0%	15.3%	42.9%	23.4%	18.97	13.62	-3.59
16	3.0b	SidewalkPilot-v3.0b.onnx	raw BGR 320x180	15.9%	14.8%	42.3%	24.6%	18.45	13.13	-3.94
17	2.3	SidewalkPilot-v2.3.pth	raw BGR	15.6%	12.8%	36.4%	34.3%	18.02	10.68	-7.22
18	2.3b	SidewalkPilot-v2.3b.pth	raw BGR	15.6%	13.0%	36.5%	32.9%	18.14	10.90	-6.92
19	1.7b	SidewalkPilot-v1.7b.pth	raw BGR	15.5%	10.8%	35.3%	51.2%	16.16	7.61	-6.82
20	2.2b	SidewalkPilot-v2.2b.pth	raw BGR	15.5%	12.7%	37.8%	28.9%	18.75	12.24	-6.96
21	1.8	SidewalkPilot-v1.8.pth	raw BGR	15.5%	10.6%	36.1%	51.4%	15.23	7.70	-5.63
22	1.8b	SidewalkPilot-v1.8b.pth	raw BGR	15.4%	10.6%	35.9%	51.9%	15.20	7.59	-5.71
23	2.2	SidewalkPilot-v2.2.pth	raw BGR	15.4%	12.6%	37.7%	29.4%	18.65	12.14	-6.72
24	1.7	SidewalkPilot-v1.7.pth	raw BGR	15.2%	10.6%	35.3%	51.1%	16.17	7.64	-6.66
25	1.1	SidewalkPilot-v1.1.pth	raw BGR	14.8%	11.0%	35.2%	45.1%	17.72	9.23	-6.49
26	1.1b	SidewalkPilot-v1.1b.pth	raw BGR	14.8%	11.0%	35.2%	45.1%	17.72	9.23	-6.49
27	1.2b	SidewalkPilot-v1.2b.pth	raw BGR	14.8%	9.9%	34.9%	51.7%	16.85	8.10	-6.77
28	1.3	SidewalkPilot-v1.3.pth	raw BGR	14.7%	9.6%	34.9%	53.9%	16.19	7.59	-6.66
29	1.6b	SidewalkPilot-v1.6b.pth	raw BGR	14.7%	10.2%	34.2%	50.1%	16.54	8.00	-5.82
30	2.4b	SidewalkPilot-v2.4b.pth	raw BGR	14.7%	12.1%	35.8%	35.3%	17.87	10.18	-7.07
31	1.3b	SidewalkPilot-v1.3b.pth	raw BGR	14.7%	9.4%	34.3%	54.8%	16.02	7.39	-6.88
32	1.6	SidewalkPilot-v1.6.pth	raw BGR	14.7%	10.2%	34.1%	49.7%	16.54	8.03	-5.87
33	2.4	SidewalkPilot-v2.4.pth	raw BGR	14.6%	12.2%	35.7%	35.0%	17.93	10.21	-7.33
34	1.2	SidewalkPilot-v1.2.pth	raw BGR	14.6%	9.6%	35.0%	52.7%	16.70	7.98	-6.55
35	1.5	SidewalkPilot-v1.5.pth	raw BGR	14.3%	8.2%	31.7%	62.1%	15.87	6.24	-6.79
36	1.5b	SidewalkPilot-v1.5b.pth	raw BGR	14.3%	8.2%	31.7%	62.1%	15.87	6.24	-6.79
37	1.4b	SidewalkPilot-v1.4b.pth	raw BGR	14.3%	9.1%	32.4%	57.3%	16.49	6.91	-6.93
38	1.4	SidewalkPilot-v1.4.pth	raw BGR	13.9%	8.6%	31.8%	57.6%	16.35	6.93	-7.31
39	1.0	SidewalkPilot-v1.0.pth	raw BGR	13.2%	10.5%	31.6%	17.0%	25.66	19.47	-14.44
40	1.0b	SidewalkPilot-v1.0b.pth	raw BGR	13.1%	10.4%	30.8%	17.6%	25.47	19.18	-14.96

Chronological Model Growth

Model	Checkpoint filename	Series	Prep	Scale	MAE	Median	Signed	<=2	<=5	<=20	Pred mean
1.0	SidewalkPilot-v1.0.pth	1	raw BGR	86	25.66	19.47	-14.44	4574	11416	41523	82.69
1.0b	SidewalkPilot-v1.0b.pth	1	raw BGR	86	25.47	19.18	-14.96	4667	11689	42055	82.16
1.1	SidewalkPilot-v1.1.pth	1	raw BGR	86	17.72	9.23	-6.49	11792	26857	57413	90.63
1.1b	SidewalkPilot-v1.1b.pth	1	raw BGR	86	17.72	9.23	-6.49	11792	26857	57413	90.63
1.2	SidewalkPilot-v1.2.pth	1	raw BGR	86	16.70	7.98	-6.55	14475	30812	59012	90.57
1.2b	SidewalkPilot-v1.2b.pth	1	raw BGR	86	16.85	8.10	-6.77	14409	30309	58787	90.35
1.3	SidewalkPilot-v1.3.pth	1	raw BGR	86	16.19	7.59	-6.66	15355	31469	59923	90.46
1.3b	SidewalkPilot-v1.3b.pth	1	raw BGR	86	16.02	7.39	-6.88	15181	31906	60463	90.24
1.4	SidewalkPilot-v1.4.pth	1	raw BGR	86	16.35	6.93	-7.31	15393	33241	59982	89.81
1.4b	SidewalkPilot-v1.4b.pth	1	raw BGR	86	16.49	6.91	-6.93	15088	33132	59757	90.19
1.5	SidewalkPilot-v1.5.pth	1	raw BGR	86	15.87	6.24	-6.79	17075	35588	60321	90.33
1.5b	SidewalkPilot-v1.5b.pth	1	raw BGR	86	15.87	6.24	-6.79	17075	35588	60321	90.33
1.6	SidewalkPilot-v1.6.pth	1	raw BGR	86	16.54	8.03	-5.87	13504	29252	59930	91.25
1.6b	SidewalkPilot-v1.6b.pth	1	raw BGR	86	16.54	8.00	-5.82	13598	29450	59968	91.30
1.7	SidewalkPilot-v1.7.pth	1	raw BGR	86	16.17	7.64	-6.66	13420	30224	60074	90.46
1.7b	SidewalkPilot-v1.7b.pth	1	raw BGR	86	16.16	7.61	-6.82	13331	30329	60092	90.30
1.8	SidewalkPilot-v1.8.pth	1	raw BGR	86	15.23	7.70	-5.63	13905	30333	61481	91.50
1.8b	SidewalkPilot-v1.8b.pth	1	raw BGR	86	15.20	7.59	-5.71	14038	30624	61544	91.41
1.9	SidewalkPilot-v1.9.pth	1	raw BGR	86	15.07	7.33	-6.09	14311	31325	61987	91.03
1.9b	SidewalkPilot-v1.9b.pth	1	raw BGR	86	15.05	7.27	-6.18	14362	31511	62036	90.94
2.0	SidewalkPilot-v2.0.pth	2	HSV/CLAHE -> BGR	85	15.52	7.80	-6.04	12289	28869	61600	91.08
2.0b	SidewalkPilot-v2.0b.pth	2	HSV/CLAHE -> BGR	85	15.48	7.74	-6.03	12374	29059	61637	91.09
2.1	SidewalkPilot-v2.1.pth	2	raw BGR	85	17.32	10.39	-6.55	9589	22710	58262	90.57
2.1b	SidewalkPilot-v2.1b.pth	2	raw BGR	85	17.23	10.32	-6.44	9696	22847	58432	90.68
2.2	SidewalkPilot-v2.2.pth	2	raw BGR	85	18.65	12.14	-6.72	7623	18630	55716	90.40
2.2b	SidewalkPilot-v2.2b.pth	2	raw BGR	85	18.75	12.24	-6.96	7486	18415	55491	90.16
2.3	SidewalkPilot-v2.3.pth	2	raw BGR	85	18.02	10.68	-7.22	8802	21430	57497	89.90
2.3b	SidewalkPilot-v2.3b.pth	2	raw BGR	85	18.14	10.90	-6.92	8463	20719	57313	90.20

Model	Checkpoint filename	Series	Prep	Scale	MAE	Median	Signed	<=2	<=5	<=20	Pred mean
2.4	SidewalkPilot-v2.4.pth	2	raw BGR	85	17.93	10.21	-7.33	9016	21932	58055	89.79
2.4b	SidewalkPilot-v2.4b.pth	2	raw BGR	85	17.87	10.18	-7.07	9093	22021	58133	90.05
3.0	SidewalkPilot-v3.0.onnx	3	raw BGR 320x180	-	18.97	13.62	-3.59	6533	16039	53679	93.53
3.0b	SidewalkPilot-v3.0b.onnx	3	raw BGR 320x180	-	18.45	13.13	-3.94	6762	16621	55133	93.18
3.1	SidewalkPilot-v3.1.onnx	3	raw BGR 320x180	-	22.65	9.73	-2.35	28493	33498	50477	94.77
3.1b	SidewalkPilot-v3.1b.onnx	3	raw BGR 320x180	-	20.96	9.59	-3.77	29744	34370	51732	93.35
3.2	SidewalkPilot-v3.2.onnx	3	raw BGR 320x180	-	16.78	9.65	-1.47	28795	33405	55311	95.65
3.2b	SidewalkPilot-v3.2b.onnx	3	raw BGR 320x180	-	14.46	5.91	-4.69	35019	39659	59670	92.43
3.3	SidewalkPilot-v3.3.onnx	3	raw BGR 320x180	-	14.63	5.20	-5.35	35471	40392	58495	91.77
3.3b	SidewalkPilot-v3.3b.onnx	3	raw BGR 320x180	-	16.46	2.18	-8.29	39671	43770	57789	88.83
3.4	SidewalkPilot-v3.4.onnx	3	raw BGR 320x180	-	14.38	4.86	0.80	35152	40809	59702	97.92
3.4b	SidewalkPilot-v3.4b.onnx	3	raw BGR 320x180	-	13.90	2.68	-2.09	38934	43481	60361	95.03

Field-Case / Subset MAE

Lower is better. Every model is evaluated on every listed Series 3 run; '-' would indicate missing results.

Model	D0702_16	D0707_16	D0712_15	D0712_16
3.2	14.46	13.64	23.37	22.43
3.4	11.69	13.02	17.00	20.73
3.1	16.00	23.15	35.18	36.67
3.1b	14.18	21.84	33.49	35.17
3.3	11.18	13.84	24.05	21.94
3.4b	11.28	15.94	17.65	19.02
3.2b	11.70	14.37	22.00	20.13
3.3b	11.27	16.20	23.50	27.85
2.0	13.38	15.81	24.45	19.55
2.0b	13.34	15.80	24.44	19.48
1.9	12.95	16.07	24.60	18.76
1.9b	12.94	16.04	24.65	18.69
2.1	15.50	18.22	23.83	20.65
2.1b	15.40	18.09	23.76	20.57
3.0	17.49	16.97	24.27	22.49
3.0b	16.85	17.06	24.90	21.94
2.3	15.75	18.82	26.23	22.22
2.3b	15.91	18.82	26.12	22.33
1.7b	13.50	17.58	26.06	20.95
2.2b	17.09	18.53	25.23	22.03
1.8	13.26	15.78	23.82	18.79
1.8b	13.21	15.74	23.83	18.79
2.2	16.94	18.59	25.13	22.00
1.7	13.56	17.56	26.07	20.82
1.1	14.80	18.44	27.48	23.31
1.1b	14.80	18.44	27.48	23.31
1.2b	13.84	18.22	27.38	22.37

Model	D0702_16	D0707_16	D0712_15	D0712_16
1.3	13.44	16.91	27.01	21.29
1.6b	13.50	17.26	26.82	22.38
2.4b	14.92	18.61	26.56	23.63
1.3b	13.36	16.45	26.60	21.04
1.6	13.55	17.27	26.73	22.25
2.4	14.97	18.59	26.68	23.74
1.2	13.75	17.92	27.04	22.17
1.5	12.95	17.33	25.84	21.22
1.5b	12.95	17.33	25.84	21.22
1.4b	13.29	17.54	27.56	22.49
1.4	13.41	17.45	27.28	21.77
1.0	23.20	25.18	36.50	30.47
1.0b	23.18	24.77	35.57	30.03

Prediction Distribution

Model	Pred min	P05	P25	Median	P75	P95	Pred max	Pred mean	Target mean
3.2	0.30	67.36	80.51	90.15	100.10	155.73	179.83	95.65	97.12
3.4	1.36	69.15	88.74	90.13	99.63	151.84	179.54	97.92	97.12
3.1	0.74	19.83	87.42	89.79	111.08	159.16	179.25	94.77	97.12
3.1b	0.95	21.03	87.67	89.94	100.39	157.76	179.06	93.35	97.12
3.3	0.66	67.78	88.57	90.10	91.51	151.77	178.73	91.77	97.12
3.4b	5.13	68.57	88.86	89.97	91.27	149.68	178.76	95.03	97.12
3.2b	0.96	68.22	88.79	90.21	91.88	129.43	178.79	92.43	97.12
3.3b	5.27	26.21	88.79	89.87	91.17	111.87	175.51	88.83	97.12
2.0	5.54	71.90	85.41	91.27	96.74	108.94	174.96	91.08	97.12
2.0b	5.55	72.05	85.47	91.29	96.71	108.79	174.96	91.09	97.12
1.9	4.01	73.10	86.10	91.14	96.12	108.25	175.97	91.03	97.12
1.9b	4.00	73.08	86.08	91.08	96.01	107.82	175.97	90.94	97.12
2.1	5.38	67.00	82.60	90.43	98.42	114.12	174.93	90.57	97.12
2.1b	5.40	67.25	82.80	90.58	98.46	114.00	174.92	90.68	97.12
3.0	9.30	65.26	81.73	93.48	104.93	122.53	172.37	93.53	97.12
3.0b	12.34	66.28	82.19	93.31	104.22	119.72	170.13	93.18	97.12
2.3	5.00	66.30	81.25	89.67	97.72	115.62	175.00	89.90	97.12
2.3b	5.00	66.30	81.31	89.85	98.29	116.30	175.00	90.20	97.12
1.7b	4.01	70.61	84.84	89.85	95.57	110.91	175.99	90.30	97.12
2.2b	5.01	62.57	80.59	90.09	100.00	117.22	174.95	90.16	97.12
1.8	5.39	73.69	86.25	91.20	96.74	109.76	175.97	91.50	97.12
1.8b	5.21	73.79	86.21	91.10	96.58	109.52	175.97	91.41	97.12
2.2	5.01	62.87	81.00	90.41	100.15	117.22	174.94	90.40	97.12
1.7	4.02	70.87	84.95	89.98	95.71	111.16	176.00	90.46	97.12
1.1	4.01	65.87	84.08	89.66	96.70	117.58	175.99	90.63	97.12
1.1b	4.01	65.87	84.08	89.66	96.70	117.58	175.99	90.63	97.12
1.2b	4.01	67.36	85.10	90.62	95.33	114.13	176.00	90.35	97.12
1.3	4.00	70.05	85.76	90.19	95.64	111.23	176.00	90.46	97.12

Model	Pred min	P05	P25	Median	P75	P95	Pred max	Pred mean	Target mean
1.6b	4.02	71.57	86.20	91.25	96.76	110.58	176.00	91.30	97.12
2.4b	5.05	66.65	82.01	90.08	97.74	113.75	175.00	90.05	97.12
1.3b	4.00	71.10	85.63	89.93	95.15	109.45	176.00	90.24	97.12
1.6	4.03	71.51	86.06	91.16	96.74	110.78	176.00	91.25	97.12
2.4	5.04	66.38	81.70	89.83	97.52	113.44	175.00	89.79	97.12
1.2	4.04	67.76	85.50	90.79	95.39	114.08	176.00	90.57	97.12
1.5	4.01	73.98	86.12	89.84	94.00	108.85	175.99	90.33	97.12
1.5b	4.01	73.98	86.12	89.84	94.00	108.85	175.99	90.33	97.12
1.4b	4.00	71.79	85.31	89.85	94.30	110.66	176.00	90.19	97.12
1.4	4.01	72.01	85.18	89.67	94.06	108.56	176.00	89.81	97.12
1.0	6.98	43.85	67.31	82.86	98.08	120.43	174.75	82.69	97.12
1.0b	7.19	45.03	67.40	82.17	96.77	118.77	174.78	82.16	97.12

Prediction Buckets vs Ground

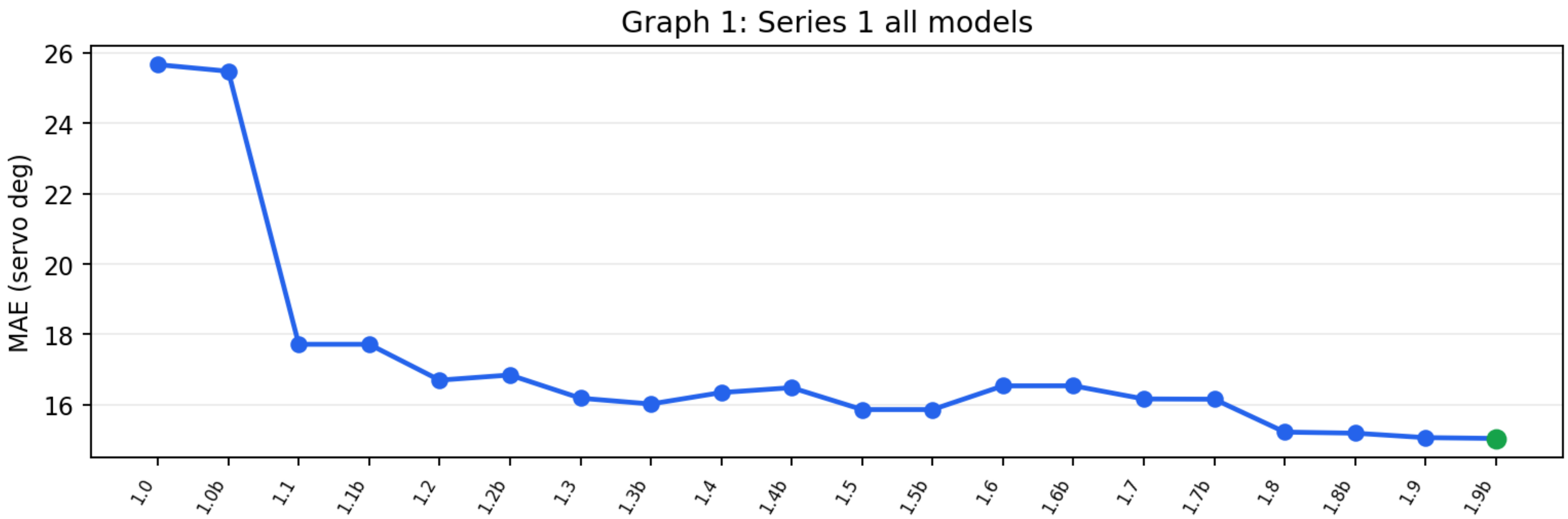
Bucket columns show prediction counts; exact/off-by-one compare prediction bucket against ground bucket.

Model	HL 0-45	L 45-75	SL 75-85	ST 85-95	SR 95-105	R 105-135	HR 135-180	Exact	Off by <=1
3.2	1947	8571	10447	34948	7681	8398	9245	46.9%	72.1%
3.4	647	4163	10475	44075	2656	9437	9784	56.3%	77.2%
3.1	7128	4482	8105	37535	3366	8513	12108	47.1%	67.1%
3.1b	6496	4504	8586	38860	2793	10380	9618	48.1%	68.2%
3.3	1649	7881	7052	48527	5328	6101	4699	54.0%	74.1%
3.4b	548	4956	8847	52257	363	7193	7073	58.2%	76.7%
3.2b	2031	3326	10037	47739	6393	7651	4060	52.8%	76.4%
3.3b	5114	2745	2420	58965	4711	4430	2852	56.2%	72.2%
2.0	434	5331	13491	35732	19933	5630	686	36.8%	75.1%
2.0b	421	5289	13346	36040	19899	5573	669	37.1%	75.2%
1.9	339	4714	12347	39785	18160	5442	450	39.8%	75.4%
1.9b	336	4704	12452	40057	18005	5251	432	40.0%	75.5%
2.1	648	8635	16273	27705	17640	9452	884	30.1%	68.9%

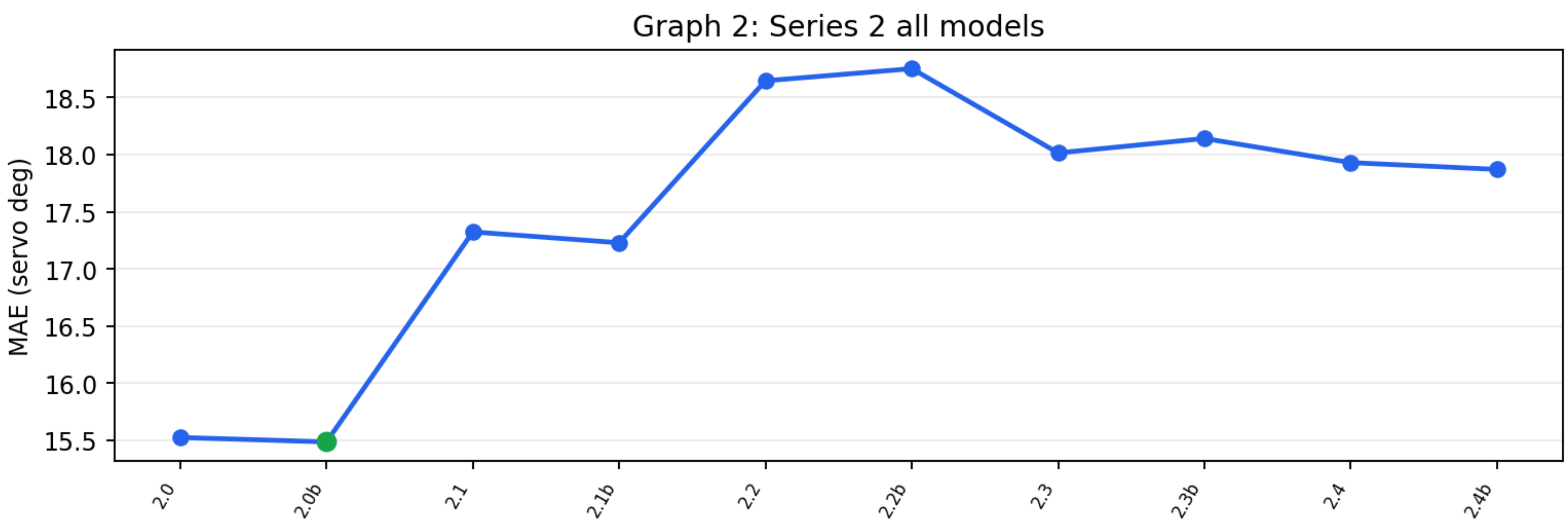
Model	HL 0-45	L 45-75	SL 75-85	ST 85-95	SR 95-105	R 105-135	HR 135-180	Exact	Off by <=1
2.1b	616	8442	15994	27894	17972	9475	844	30.2%	69.1%
3.0	325	10946	14167	18078	17519	19277	925	23.0%	61.7%
3.0b	265	10415	14295	18938	18306	18384	634	23.5%	63.1%
2.3	655	9931	17905	26197	16776	8782	991	28.3%	67.5%
2.3b	646	9939	17722	25306	17171	9378	1075	27.5%	67.1%
1.7b	640	5329	14858	38573	15094	5889	854	38.4%	73.1%
2.2b	978	11590	16414	22419	16883	11875	1078	25.1%	64.4%
1.8	215	4506	12179	38927	18371	6679	360	38.7%	74.6%
1.8b	213	4444	12304	39305	18145	6470	356	38.9%	74.7%
2.2	963	11132	16062	22757	17314	11954	1055	25.4%	64.8%
1.7	636	5226	14601	38561	15374	5956	883	38.3%	73.1%
1.1	1082	6803	14941	34540	14026	8318	1527	34.7%	69.0%
1.1b	1082	6803	14941	34540	14026	8318	1527	34.7%	69.0%
1.2b	1052	6359	12695	39883	13774	6296	1178	38.5%	70.8%
1.3	923	5029	12087	41359	14520	6681	638	39.9%	72.5%
1.6b	1004	4439	11780	38345	18904	5701	1064	37.4%	72.5%
2.4b	744	8624	17750	26944	18179	8060	936	28.6%	68.9%
1.3b	796	4759	12656	42347	14356	5657	666	40.3%	73.1%
1.6	982	4517	12108	38082	18712	5787	1049	37.1%	72.4%
2.4	756	8903	18227	26755	17785	7893	918	28.5%	68.7%
1.2	990	6129	12142	40532	14000	6271	1173	39.1%	71.1%
1.5	800	3658	11722	47843	11723	4534	957	44.5%	72.8%
1.5b	800	3658	11722	47843	11723	4534	957	44.5%	72.8%
1.4b	875	4513	13939	43971	12264	4345	1330	41.7%	72.1%
1.4	840	4523	14323	44378	11974	4183	1016	41.7%	72.2%
1.0	4486	25232	13937	13523	10904	12025	1130	16.7%	46.5%
1.0b	4055	26108	14623	13953	10776	10640	1082	16.9%	46.8%

Graphs

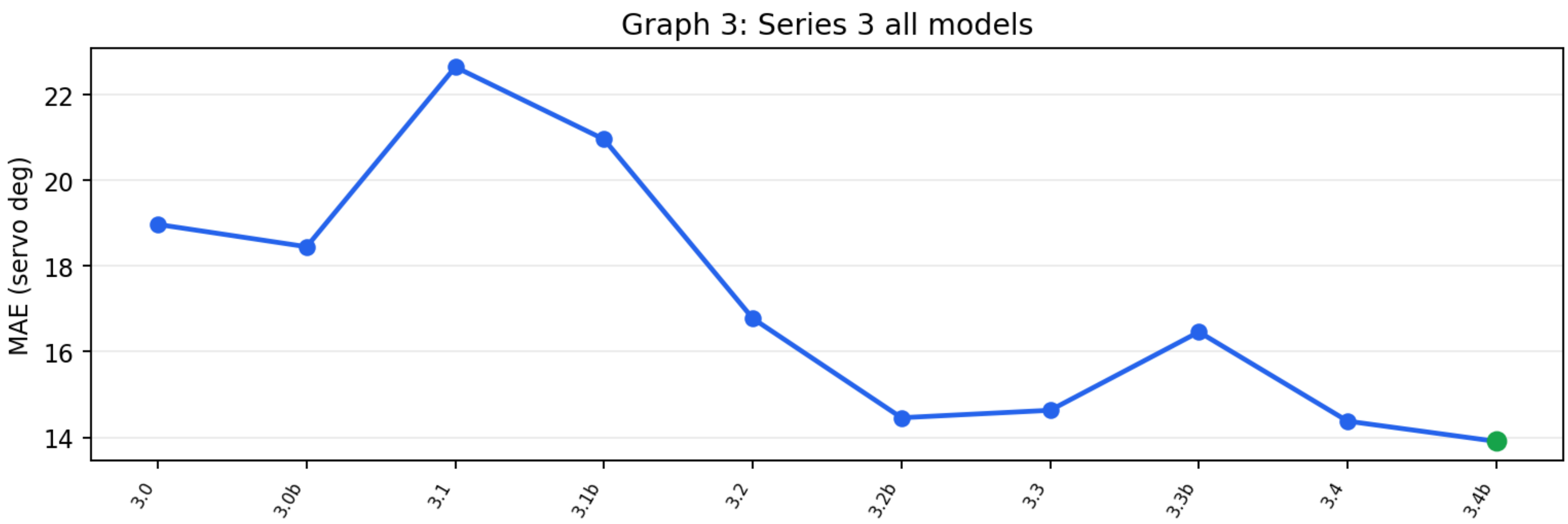
Graph 1: Series 1 all models



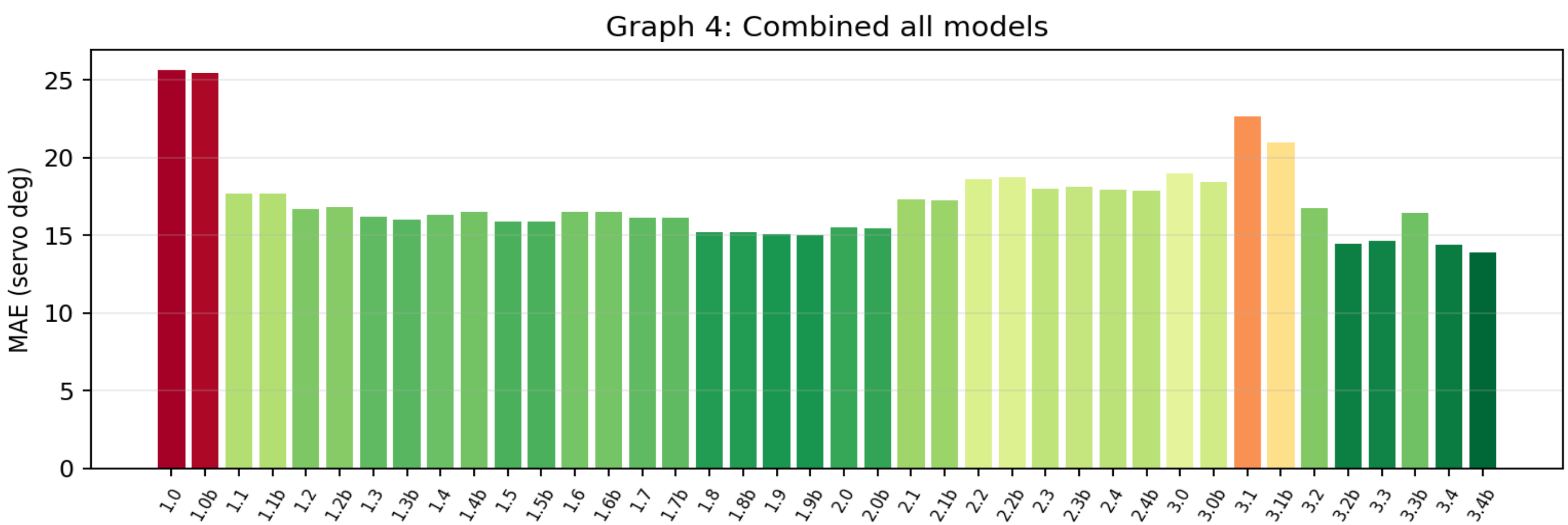
Graph 2: Series 2 all models



Graph 3: Series 3 all models



Graph 4: Combined all models



Steering-Bucket Confusion (9-class) — latest 4 models

Rows = the TRUE steering bucket, columns = the model's PREDICTED bucket. Green diagonal = correct; red off-diagonal = confusion. Shading = share of that true bucket (each row sums to ~100%). Buckets: HL 0-45, L 45-60, L+ 60-75, SL 75-85, ST 85-95, SR 95-105, R 105-120, R+ 120-135, HR 135-180.

v3.4b · exact bucket 57% · within one bucket 74%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	146	34	220	81	152	1	2	2	53
L	28	36	362	218	363	1	2	2	15
L+	28	47	1043	671	842	3	6	9	34
SL	43	49	817	1510	1680	2	11	29	58
ST	133	120	2061	5852	39710	160	1340	1588	2946
SR	6	1	27	114	1746	52	259	246	344
R	6	1	23	138	2315	33	461	397	525
R+	8	·	32	77	2068	41	557	569	709
HR	150	5	78	186	3381	70	986	727	2389

v3.4 · exact bucket 54% · within one bucket 74%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	260	27	155	109	129	1	3	1	6
L	19	85	374	243	289	3	4	·	10
L+	21	27	1089	860	645	11	7	4	19
SL	33	10	643	2208	1223	19	21	12	30
ST	144	67	1636	6739	35306	1283	2867	1467	4401
SR	1	·	11	72	1301	344	408	239	419
R	4	·	6	69	1557	275	905	373	710
R+	5	·	4	42	1336	311	711	746	906
HR	160	·	29	133	2289	409	1149	520	3283

v3.3b · exact bucket 55% · within one bucket 70%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	245	·	110	40	244	19	14	·	19
L	176	1	154	66	573	40	12	·	5
L+	427	3	440	217	1413	128	36	·	19

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
SL	440	2	387	389	2669	211	70	1	30
ST	3071	18	1326	1457	42065	3051	1820	99	1003
SR	111	·	40	44	1961	219	288	10	122
R	146	·	56	51	2726	280	418	19	203
R+	145	·	56	41	2604	294	565	40	316
HR	353	·	152	115	4710	469	979	59	1135

v3.3 · exact bucket 53% · within one bucket 71%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	300	34	132	42	156	15	6	·	6
L	76	67	347	109	370	32	10	3	13
L+	136	90	1010	431	853	102	50	4	7
SL	149	120	844	1205	1587	182	99	4	9
ST	757	795	3502	4578	36908	3256	2239	349	1526
SR	20	17	108	117	1596	333	307	92	205
R	28	49	181	144	2066	375	518	187	351
R+	30	23	162	134	1878	387	562	337	548
HR	153	60	340	292	3113	646	986	348	2034

Notes

- Every checkpoint was evaluated on the same 81,237 Series 3 images. Series 2 v2.0/v2.0b used their required HSV/CLAHE preprocessing; the other Series 1/2 models used raw BGR.
- The dataset is 66% straight, so MAE and within-degree counts can reward straight collapse. Use the class-balanced and turn-recall columns for selection, with MAE as supporting evidence.
- Offline MAE does not prove real-world reliability. The car can still fail on lighting, turns, driveways, road-edge ambiguity, speed, and sensor conditions.
- Series 1/2 were trained on older real-plus-CARLA datasets, so this is an out-of-era transfer test for them rather than their original validation benchmark.
- Series 3 models were trained on overlapping images from this dataset, so their scores are fit checks, not held-out generalization estimates.
- The Series 3 evaluation set contains 81,237 curated real field frames across five manual-driving runs from July 2 through July 12, 2026.
- Series 3 throttle labels are near-constant at full throttle, so this report evaluates steering only; throttle control remains disabled pending varied-throttle data.
- v3.3, v3.3b, v3.4, and v3.4b still require field testing before any deployment winner is declared.